

JOSEPH KURIAN JOHN
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Education

Delhi Technological University (Formerly DCE), India

2022 - 2026

- Branch: Electrical Engineering | CGPA: 8.5

Skills

C++ | Embedded C | Python | Basic Java | UART | GPIO | Timers | FreeRTOS| ESP32 | Git

Work Experience

GenElek Technologies| Embedded Electronics Intern

May'26 - Present| C++, MAB CANDle-SDK, BLDC Motors, Control Methodologies

- Optimized communication performance and timing consistency across a high-speed FDCAN bus by implementing non-blocking asynchronous writing and reading methods within the MAB API, ensuring synchronized trajectory execution and joint feedback updates for daisy-chained actuators with millisecond latency.
- Developed a safety-critical monitoring and system diagnostic module that decoded real-time motor status bitmasks and power rails to dynamically flag erroneous conditions, initiating automated software-defined shut-off routines to guarantee patient safety during gait-training sessions.
- Engineered a real-time multi-joint control architecture for a clinical-grade robotic physiotherapy machine by integrating MAB Robotics MD80 brushless controllers and configuring active impedance control (PD mode) loops, providing safe, compliant joint assistance tailored to the varying physical capacities of stroke and spinal cord injury patients.

Projects

[Satellite Telemetry Anomaly Detection \(HIL Simulation\)](#)

Hardware-in-the-Loop (HIL) | Python, C++, TensorFlow Lite Micro, ESP32

- Integrated Hardware & Software: Engineered a Python-based simulation environment that streamed real-time synthetic satellite health data (voltage, temperature) to an ESP32 microcontroller via Serial communication.
- Edge AI Deployment: Developed and trained a Tiny Autoencoder in Python, optimized the model using TensorFlow Lite for Microcontrollers, and deployed it onto the ESP32 for real-time inference.
- Real-time Anomaly Detection: Programmed the embedded system to perform on-device signal processing, identifying data "spikes" and anomalies with low latency and triggering automated alerts back to the host system.

[Edge AI Speech Processing & Hardware Simulation Pipeline \(Cochlear Implant Prototype\)](#)

Edge Computing | C++, Python, Scikit-Learn, Signal Processing

- Architected a closed-loop Hardware-in-the-Loop (HIL) system deploying a quantized CNN on an ESP32 for ultra-low latency (<15ms) MFCC audio classification, coupled with Python-based virtual oscilloscope simulations modeling RC filter electrical stimulation dynamics.

[High-Performance Parallelized Fractal Engine | ESP32 SoC](#)

- Engineered a parallel processing framework for the ESP32 (Xtensa Dual-Core) to compute complex Mandelbrot set iterations, utilizing Symmetric Multiprocessing (SMP) to achieve a near 2x increase in computational throughput.
- Optimized task distribution by implementing a custom load-balancing algorithm, ensuring maximum utilization of both CPU cores while managing FreeRTOS task priorities and resource contention.

- Developed a Python-based visualization interface to ingest and render the incoming data stream.

Positions of Responsibility

AIMS DTU | Member Jun'22 - Jun'23

- Collaborated on software and machine learning projects in a team-driven environment, contributing to innovation and practical implementation. Mentored junior students by providing technical guidance and support to help them develop impactful projects using programming and basic ML concepts.